

MODULE 5: INTERNET SECURITY PROTOCOL, USER AUTHENTICATION

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Internet Security Protocols

Internet Security Protocols are a collection of rules and standards used to protect data transmitted across the Internet and computer networks. They ensure that communication remains confidential, authentic, and free from unauthorized modification. **Need for Internet Security Protocols**

1. Protect sensitive information during transmission.
2. Prevent unauthorized access to network resources.
3. Ensure confidentiality of communication.
4. Maintain data integrity.
5. Provide authentication of communicating parties.
6. Support secure e-commerce and online banking.

Security Services Provided

1. **Confidentiality:** Protects information from unauthorized disclosure through encryption.
2. **Integrity:** Ensures that transmitted data is not modified by attackers.
3. **Authentication:** Verifies the identity of communicating entities.
4. **Non-Repudiation:** Prevents a sender from denying that a message was sent.
5. **Access Control:** Restricts resource access to authorized users only.

Common Internet Security Protocols

- | | | |
|-----------|---------|----------|
| • SSL/TLS | • IPSec | • PGP |
| • HTTPS | • SSH | • S/MIME |

SSL (Secure Socket Layer) Protocol

SSL is a cryptographic protocol developed for securing communication over the Internet. Modern systems use TLS (Transport Layer Security), which is the successor to SSL. SSL provides:

- | | | |
|------------------|-------------------|-------------|
| • Authentication | • Confidentiality | • Integrity |
|------------------|-------------------|-------------|

Working Principle: SSL establishes a secure connection between a client and server before data transmission begins.

SSL certificates

An SSL Certificate is a bit of code on our web server that provides security for online communications. They safeguard sensitive user data from interception and allow browsers to verify that they are communicating with the legitimate server rather than an imposter. SSL operates right between the Application layer and the Transport layer of the OSI network model

Why You Need Them

- **Encryption:** They facilitate SSL/TLS encryption, turning unreadable internet traffic into secure code.
- **Trust & Verification:** Browsers display a padlock icon and "HTTPS" to reassure users that the connection is safe.
- **Search Engine Optimization (SEO):** Major search engines give ranking advantages to HTTPS websites.

How They Work

SSL certificates contain a website's public key, domain details, and a digital signature from a trusted Certificate Authority (CA). When a user connects to a secured site, the browser verifies this signature. Once verified, it creates a secure session, safeguarding any information (like login credentials or credit card numbers) transferred between the user and the server.

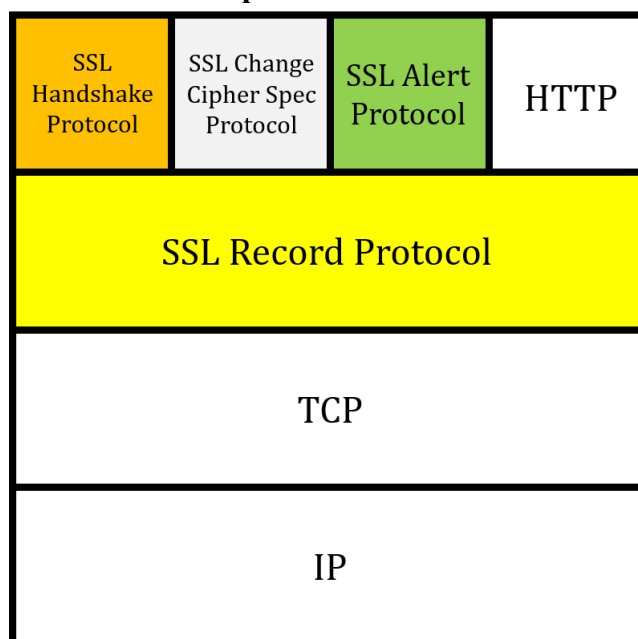
SSL Architecture

SSL acts as an intermediary layer between application protocols (such as HTTP) and the TCP/IP protocol suite, providing secure communication over the Internet. **Purpose of SSL:**

1. Confidentiality through encryption.
 2. Authentication of communicating parties.
 3. Data integrity verification.
 4. Secure key exchange.
1. **Application Layer (HTTP):** HTTP is an application layer protocol used for web communication.
 - Normally, HTTP sends data in plain text.
 - When SSL is used, HTTP becomes HTTPS.
 - All data exchanged between the browser and server is protected by SSL.

Example:

- HTTP → Insecure
- HTTPS → Secure using SSL/TLS



2. **SSL Handshake Protocol:** The SSL Handshake Protocol is responsible for establishing a secure connection between the client and server.

Functions

1. Negotiates SSL/TLS version.
2. Selects encryption algorithms (Cipher Suite).
3. Authenticates the server.
4. Optionally authenticates the client.
5. Establishes session keys.

Handshake Steps

1. Client Hello
2. Server Hello
3. Certificate Exchange
4. Key Exchange
5. Session Key Generation
6. Secure Communication Begins

3. Change Cipher Spec Protocol: is a simple protocol used during the SSL handshake.

- Notifies the other party that future messages will use the negotiated security parameters.
- Activates the selected encryption and hashing algorithms.
- Marks the transition from an unsecured state to a secured state.

Characteristics

- Consists of only one message.
- Sent by both client and server.
- Very small protocol but essential for secure communication.

4. SSL Alert Protocol: The Alert Protocol is used to communicate errors and status information.

1. Reports problems during SSL communication.
2. Indicates whether a connection should continue or terminate.
3. Provides warning and fatal alerts.

- Certificate expiration warning.
- Close notification.

2) **Fatal Alert:** Critical errors requiring immediate connection termination.

Examples:

- Invalid certificate.
- Decryption failure.
- Handshake failure.

Types of Alerts

1) **Warning Alerts:** on-critical issues.
Examples:

5. SSL Record Protocol: The SSL Record Protocol is the core component of SSL. All SSL messages pass through this protocol before transmission.

Functions

1. Fragmentation of data.
2. Compression (optional).
3. Message Authentication Code (MAC) generation.
4. Encryption of data.
5. Secure transmission.

4. Encryption: Data and MAC are encrypted using the session key.

5. Transmission: Encrypted records are sent through TCP.

Importance

The SSL Record Protocol provides:

- Confidentiality
- Integrity
- Authentication

and serves as the foundation for all SSL operations.

Working

1. **Fragmentation:** Large messages are divided into smaller blocks.
2. **Compression:** Data may be compressed to reduce size.
3. **MAC Generation:** A Message Authentication Code is generated for integrity verification.

6. TCP Layer Transmission Control Protocol (TCP) provides:

- Reliable communication.
- Error detection.
- Flow control.
- Ordered delivery of packets.

SSL uses TCP as its transport mechanism.

7. IP Layer: Internet Protocol (IP) is responsible for:

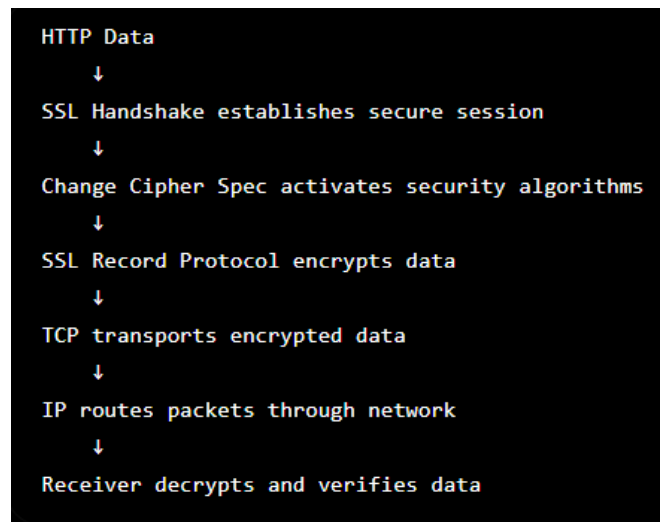
- Addressing.
- Routing packets.
- Delivering packets across networks.

SSL does not directly interact with IP; communication occurs through TCP.

Working of SSL Architecture (figure ->)

Advantages of SSL Architecture

1. Provides end-to-end security.
2. Ensures confidentiality through encryption.
3. Verifies identity using digital certificates.
4. Detects message tampering.
5. Protects against eavesdropping attacks.
6. Forms the basis of HTTPS communication.
7. Widely used in e-commerce, online banking, and secure web services.



User Authentication

User Authentication is the process of verifying the identity of a user before allowing access to a computer system, network, or application. Authentication acts as the first line of defense against unauthorized access and cyberattacks. **Objectives**

1. Verify user identity.
2. Protect confidential resources.
3. Prevent impersonation attacks.
4. Maintain accountability.
5. Support secure communication.

Authentication Process

1. User submits credentials.
2. System verifies credentials.
3. Authentication server checks validity.
4. Access is granted or denied.

Authentication mechanisms are generally classified according to the type of credential being verified.

Authentication Factors

1. **Something You Know:** Information known only to the user. Examples: Passwords, PINs, Security Questions
2. **Something You Have:** Physical objects possessed by the user. Examples: Smart Cards, Authentication Tokens, Mobile Phones
3. **Something You Are:** Unique biological characteristics. Examples: Fingerprints, Face Recognition, Iris Recognition
4. **Something You Do:** Behavioral characteristics. Examples: Signature Verification, Typing Rhythm, Voice Pattern
5. **Somewhere You Are** Location-based authentication. Examples: GPS Location, Network Address

Multi-Factor Authentication (MFA)

MFA combines two or more authentication factors to provide stronger security. Example: Password + OTP/Fingerprint Benefits:

1. Increased security.
2. Reduced risk of credential theft.
3. Better protection against phishing attacks.

Biometric Authentication

Biometric Authentication identifies individuals based on unique physical or behavioral characteristics. Because biometric traits are unique to each person, they provide a high level of security.

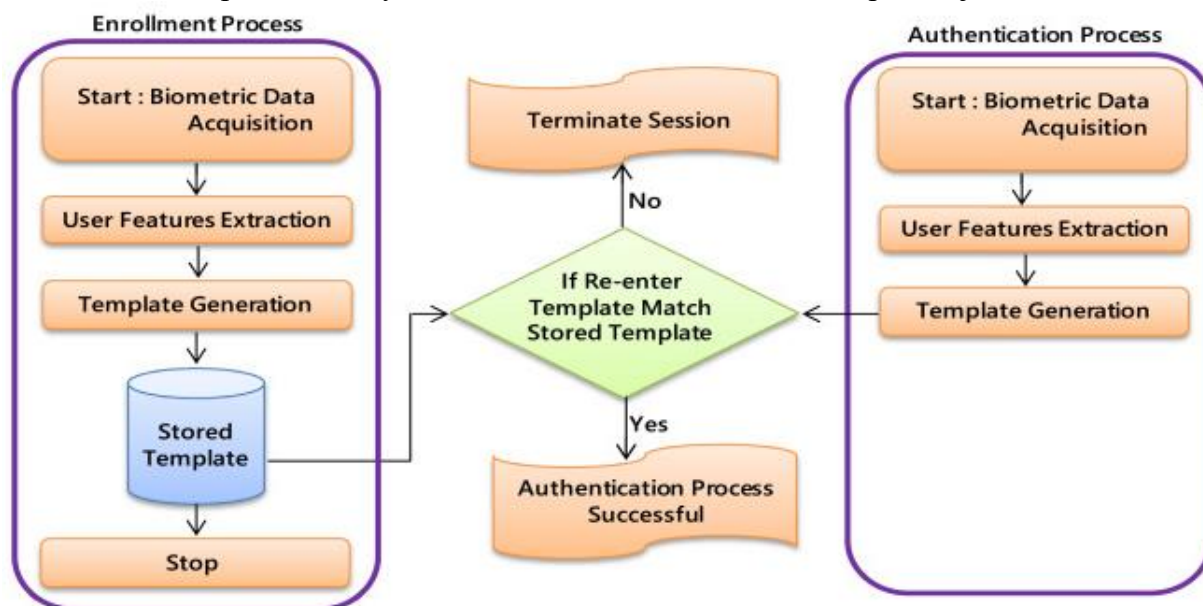
Biometric Authentication Process

1) Enrolment Phase

- Capture biometric sample.
- Extract unique features.
- Create biometric template.
- Store template securely.

2) Authentication Phase

- Capture new biometric sample.
- Compare with stored template.
- Calculate matching score.
- Accept or reject user.



Types of Biometrics

1) Physiological Biometrics: Based on physical traits. Examples:

- Fingerprint
- Face Recognition
- Iris Recognition
- Retina Scan
- Palm Print

2) Behavioral Biometrics Based on user behavior. Examples:

- Voice Pattern
- Typing Rhythm
- Signature Dynamics
- Gait Recognition

Performance Measures

1. **FAR (False Acceptance Rate):** Measures the probability of unauthorized users being accepted.
2. **FRR (False Rejection Rate):** Measures the probability of legitimate users being rejected.
3. **EER (Equal Error Rate):** The point where FAR equals FRR. Lower EER indicates a more accurate biometric system.

Advantages

1. Very high security.
2. Convenient to use.
3. Cannot be forgotten.
4. Difficult to duplicate.

Disadvantages

1. Privacy concerns.
2. Expensive devices.
3. False acceptance and rejection.
4. Biometrics cannot be changed if compromised.

Password Authentication

Password Authentication is the most widely used authentication mechanism where a user proves identity by providing a secret password. **Working**

1. User enters username.
2. User enters password.
3. Password hash is generated.
4. Hash is compared with stored hash.
5. Access granted if matched.

Password Hashing: Instead of storing passwords directly, systems store hash values.

Common Algorithms:

- SHA-256
- bcrypt
- PBKDF2
- Argon2

Hashing protects passwords even if the database is compromised.

Characteristics of Strong Passwords

1. Minimum 8–12 characters.
2. Combination of upper and lowercase letters.
3. Includes numbers.
4. Includes special symbols.
5. Avoids dictionary words.
- 3) **Phishing Attack** Tricking users into revealing passwords.
- 4) **Shoulder Surfing:** Observing a user entering credentials.

Advantages

1. Easy to implement.
2. Low deployment cost.
3. Familiar to users.

Password Attacks

- 1) **Brute Force Attack:** Trying every possible password combination.
- 2) **Dictionary Attack:** Using a list of common passwords.

Disadvantages

1. Easy to forget.
2. Vulnerable to attacks.
3. Password reuse problems.

Authentication Token

An Authentication Token is a physical or software-generated device that produces temporary authentication credentials. Tokens are commonly used in Two-Factor Authentication (2FA).

Types of Authentication Tokens

- 1) **Hardware Tokens:** Physical devices that generate OTPs. Examples: RSA SecureID, Smart Cards
- 2) **Software Tokens** Applications generating OTPs. Examples: Google Authenticator, Microsoft Authenticator
- 3) **SMS Tokens** One-Time Passwords delivered via SMS.

Working of Token Authentication

1. User enters username and password.
2. User enters generated OTP.
3. Server verifies OTP.
4. Access is granted.

The OTP changes periodically, making it difficult for attackers to reuse credentials.

Advantages

1. Strong security.
2. Supports MFA.
3. Difficult to duplicate.

Disadvantages

1. Additional hardware cost.
2. Token loss issues.
3. Synchronization problems.

Certificate-Based Authentication

Certificate-Based Authentication verifies identity using a digital certificate issued by a trusted Certificate Authority (CA). It relies on Public Key Infrastructure (PKI).

Digital Certificate: A digital certificate contains:

- User identity
- Serial number
- Validity period
- Public key
- Issuer information
- CA signature

X.509 Certificate: X.509 is the international standard format for digital certificates.

Diagram: X.509 Certificate Structure

Public Key Infrastructure (PKI): PKI is a framework that manages digital certificates and public keys. **Components**

- 1) **Certificate Authority (CA):** Issues digital certificates.
- 2) **Registration Authority (RA):** Verifies user identities.
- 3) **Certificate Repository:** Stores certificates.
- 4) **End Users:** Use certificates for authentication.

Diagram: PKI Architecture

Certificate Authentication Process

1. User presents certificate.
2. Server validates certificate.
3. Server verifies CA signature.
4. Public key is extracted.
5. Authentication succeeds.

Applications:

- HTTPS websites
- VPNs
- Enterprise networks

- Government systems

Advantages

2. Strong authentication.
3. Difficult to forge.
4. No password management issues.

Disadvantages

1. Certificate management overhead.
2. PKI deployment cost.
3. Certificate expiration issues.